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Decreased mismatch negativity and elevated frontal-lateral connectivity in first-episode psychosis

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ABSTRACT

Decreased mismatch negativity (MMN) is a proposed biomarker for psychotic disorders. However, the magnitude of the effect appears to be attenuated in first-episode populations. Furthermore, how mismatch negativity amplitudes are related to brain connectivity in this population is unclear. In this study, we used high-density EEG to record duration-deviant MMN from 22 patients with first-episode psychosis (FEP) and 23 age-matched controls (HC). Consistent with past work, we found decreased MMN amplitude in FEP over a large area of the frontal scalp. We also found decreased latency over the occipital scalp. MMN amplitude was negatively correlated with antipsychotic dose. We used Granger causality to investigate directional connectivity between frontal, midline, left, and right scalp during MMN and found reduced connectivity in FEP compared to HC and following deviant stimuli compared to standard stimuli. FEP participants with smaller decreases in connectivity from standard to deviant stimuli had worse disorganization symptoms. On the other hand, connectivity from the front of the scalp following deviant stimuli was relatively preserved in FEP compared to controls. Our results suggest that a relative imbalance of bottom-up and top-down perceptual processing is present in the early stages of psychotic disorders.

1. Introduction

Schizophrenia and other psychotic disorders are common and severe psychiatric illnesses. Schizophrenia typically presents in young adulthood and can lead to major disruptions in academic, occupational and social achievement (Whiteford et al., 2015). The early course of schizophrenia may be a critical time for interventions to improve patient outcomes (McGorry, 1998). However, delivering effective treatments to this population is complicated by multiple factors. Current treatments for schizophrenia have limited efficacy and it is difficult to target these treatments to the population that will benefit. Both of these issues could be helped by a better understanding of the biological underpinnings of this disease. However, the pathology of schizophrenia is likely multifactorial and remains somewhat unclear. Several neurotransmitter systems have been implicated, including GABA, glutamate, dopamine, and acetylcholine (Lisman et al., 2008). A neuroimaging biomarker for psychotic disorders could be a non-invasive, easy to obtain way to identify patients and could also shed light on the pathophysiology of the disease.

One potential biomarker is mismatch negativity (MMN), an event-related potential that occurs when a sequence of repeated identical (standard) stimuli is interrupted by an odd (deviant) stimulus (Näätänen, 1995). MMN can occur in response to a variety of stimuli and deviations. MMN is an automatic, unconscious process that depends on bottom-up perceptual processing to detect deviant stimuli (Kenemans and Kähkönen, 2011). Auditory MMN paradigms are commonly employed in which either pitch or duration may be varied and are generally measured over frontocentral electroencephalographic (EEG) channels (Umbricht and Krljesb, 2005). Decreased auditory MMN has been widely reported in patients with schizophrenia and has been proposed to be a biomarker for psychotic disorders (Light and Näätänen, 2013; Umbricht and Krljesb, 2005). However, the magnitude of the decrease may be related to illness duration with first-episode patients showing no reduction in MMN to pitch deviants and small, but statistically significant reduction in MMN to duration deviants (Haigh et al., 2017). MMN impairment may reflect premorbid intellectual functioning

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in first-episode psychosis as well as a level of vulnerability to disease progression in high-risk psychosis population (Erickson et al., 2016; Salisbury et al., 2017). Furthermore, MMN deficits may not be specific to schizophrenia as they have also been reported in bipolar disorder, autism spectrum disorder, and with mixed evidence for major depressive disorder (Bissonnette et al., 2020; Hermens et al., 2018; Schwartz et al., 2018). This suggests that the neural circuits that produce MMN may be disrupted by a variety of psychiatric disorders.

Some have proposed that schizophrenia and other psychotic disorders are dysconnection syndromes (Friston and Frith, 1995). In this model, psychosis symptoms arise from aberrant functional integration across brain regions. There is a robust literature describing abnormalities in both task-related and resting state connectivity across multiple frequency bands in both high-risk, first episode and chronic populations (Perrottelli et al., 2021). Recently, altered directional functional connectivity during mismatch negativity has been reported in patients with chronic schizophrenia (Koshiyama et al., 2020). This suggests that MMN may function as a probe of dysconnectivity in this context though which features of the pathophysiology of schizophrenia contribute to decreased MMN has yet to be determined. Decreased MMN might be caused by impaired function of local cortical circuits within the auditory cortex and/or dysconnectivity across specific long-range cortico--thalamo-cortical connections implementing bottom-up processing. Furthermore, different patterns of dysconnectivity may underlie MMN impairments at different stages of illness.

In this study, we measured MMN to a duration deviant in patients with first-episode psychosis compared to healthy controls and investigated differences in brain connectivity following standard and deviant stimuli using high-density EEG. We hypothesized that, consistent with prior literature, the MMN would be present but attenuated in patients relative to controls. We also hypothesized that this difference would be related to differences in long-range cortical connectivity consistent with a relative weakening of bottom-up perceptual processing to deviant stimuli. We hypothesized that causal density, a measure of the complexity of the connectivity pattern in a network, would be abnormal in patients with first-episode psychosis consistent with the dysconnection hypothesis.

2. Methods

2.1. Participants and procedure

We recruited 30 patients with first-episode psychosis (FEP, within 3 years of initial diagnosis of schizophrenia, schizoaffective disorder, bipolar disorder with psychotic features, or psychosis not otherwise specified) from inpatient units and outpatient clinics at McLean Hospital and 30 age-matched healthy controls (HC) with no history of mental illness who were recruited through advertisements in the community. Exclusion criteria were history of major medical or neurological illness, history of severe head injury, hearing loss, and history of electroconvulsive therapy. Control participants could not have any history of use of psychiatric medication or of psychiatric diagnoses. All participants from an inpatient unit had an independent evaluation of their ability to consent to the study procedures performed by a psychiatrist not involved in the study. All participants gave informed consent for all study procedures. All study procedures were approved by the Institutional Review Board of Partners Healthcare. All patient participants were receiving treatment at McLean Hospital and had clinical diagnoses were obtained from the medical record and confirmed with the Structured Clinical Interview for DSM-5 (First et al., 2015). All patients had a brief hearing screen. FEP patient participants underwent a video-recorded Structured Clinical Interview for the Positive and Negative Symptom Scale (SCI-PANSS) (Kay et al., 1987). Raw PANSS scores were converted to factors using a weighted sum approach (DiStefano et al., 2009; Wallwork et al., 2012). After the interview, participants EEG data was collected. Two control subjects and three patients did not tolerate the experiment and ended it early and their data was not included in this analysis. Following data collection, one control subject was found to not meet inclusion criteria (use of psychiatric medications) and three patients were subsequently found to not meet the first-episode criteria. Data from four control subjects and two patients was contaminated by persistent broadband EEG artifact. Therefore, the final data set consisted of 22 patients and 23 controls. Demographic information about the sample is presented in Table 1.

2.2. Electroencephalography

EEG data collection was performed using a Geodesic Sensor Net (Electrical Geodesics Incorporated; Philips Amsterdam, The Netherlands) with 129 Ag–AgCl electrodes. EEG data were collected at a 1 kHz sampling rate with a common vertex reference using the NetStation software package (EGI). The collection took place in an electrically and acoustically shielded room. Impedances were kept below 65 kΩ. Bad channels and artifactual segments of data were identified by visual inspection and a thresholding procedure. Bad channels were interpolated by spherical splines and artifactual segments of data were excluded from the analysis. The data were then bandpass filtered from 0.5 to 40 Hz, downsampled to 250 Hz, re-referenced to linked mastoids, and exported to MATLAB (MathWorks, Natick MA).

2.3. Auditory mismatch negativity

We elicited auditory MMN using a duration deviant. Participants were seated in a comfortable chair and watched a short, silent video of nature scenes. Auditory stimuli were delivered via in-ear headphones (Etymotics, ER3C). Standard stimuli were 1000 Hz, 50 ms and deviant stimuli were 1000 Hz, 100 ms. We presented 1000 stimuli with an average interstimulus interval of 500 ms. We presented 900 standard stimuli and 100 deviant stimuli. We first presented a run of 100 standard stimuli after which point deviant stimuli were randomly interspersed. For each participant, we calculated MMN then by subtracting the mean baseline-corrected response to the standard stimulus from the mean baseline-corrected response to the deviant stimulus. Latency was calculated as the time point of maximal difference in amplitude between the standard and deviant stimuli over this interval. We calculated the MMN amplitude by taking the mean difference over a 50 ms window centered on the latency time point.We used statistical nonparametric permutation tests (SnPM) at the single channel and cluster level to identify differences in MMN between the HC and FEP groups. In channel-level SNPM, t-statistics are calculated for each channel and the largest magnitude t-value is recorded. Then group labels are permuted, and this process is repeated for 10,000 permutations. The 95% percentile t-value is used as a threshold for the t-values from the non-permuted comparison and any t-values in excess of this percentile are considered statistically significant. For SnPM cluster tests, a similar permutation analysis is performed using the size of spatially contiguous cluster of channels that all exceed a chosen t value threshold. Given that little is known about the relationship between symptom severity and connectivity during MMN paradigms in people with early course psychosis, we

Table 1
Demographic and clinical information.

	Control	Patient
Sample Size	23	22
Age (Years, Mean ± SEM)	23.2 ± 0.5	22.6 ± 0.6
Sex (Male, Female)	12,11	16,6
Chlorpromazine Equivalents (mg, Mean ± SEM)	–	208.3 ± 32.8
PANSS Positive Factor (Mean ± SEM)	–	7.6 ± 0.7
PANSS Negative Factor (Mean ± SEM)	–	10.4 ± 0.8
PANSS Disorganized Factor (Mean ± SEM)	–	4.6 ± 0.4
PANSS Excited Factor (Mean ± SEM)	–	5.3 ± 0.4
PANSS Depressed Factor (Mean ± SEM)	–	4.2 ± 0.3

performed a hypothesis-generating, exploratory analysis by calculating the Pearson correlation coefficient between MMN connectivity parameters and medication dose and symptom scores. Because these comparisons are hypothesis-generating they are not corrected for multiple comparisons.

2.4. Connectivity analyses

We used Granger causality (GC) to measure the long-range connectivity between four channels in a 350 ms window following stimulus presentation. These channels were chosen because they were located at opposite ends of a cluster that showed decreased MMN amplitude in FEP compared to HC (see below). We chose to analyze a subset of the cluster because neighboring electrodes can be highly correlated. The Granger causality from signal *A* to signal *B* is a measure of how much information about past values of *A* improves the prediction of future values of *B* beyond what can be learned from past values of *B*. In multivariate Granger causality analysis, the connectivity value for each channel is conditioned on the contributions from other channels (Barnett and Seth, 2014), Therefore, having multiple correlated channels would obscure the connectivity results. By choosing four spatially distinct channels, we minimize correlations due to volume conduction and bridging while covering the spatial extent of the cluster. We calculated the causal density of the network formed by these four channels. Causal density of a network is the average pairwise GC conditioned by the other connections (Seth et al., 2011). Causal density is a measure of the complexity of a network because a network composed of tightly correlated channels will have a low value for causal density as will a network composed of independent channels. High values of causal density indicate a complex pattern of GC between elements of a network.

3. Results

3.1. Mismatch negativity amplitude is decreased frontally and latency is decreased occipitally

For each participant, we calculated the mismatch negativity (MMN) amplitude and latency for each channel and the average MMN amplitude and latency across all channels (Fig. 1a and b). We found no statistically significant difference between mean amplitude in HC (−2.04 μV, standard error of the mean (SEM) 0.20 μV) and FEP (−1.49 SEM 0.19 μV, $p = .078$, unpaired *t*-test, Fig. 1b). We also found no statistically significant difference in mean latency between HC (183.74 SEM 3.70 ms) and FEP (182.23 SEM 5.31 ms, $p = .80$, unpaired *t*-test). We found that increased mean latency was associated with increased magnitude of the MMN in FEP (Pearson's $r = -.49$, $p = .021$) but not in HC (Fig. 1c). Within FEP, we found that CPZ equivalents were associated with decreased amplitude of the MMN (Pearson's $r = 0.51$, $p = .015$) and with decreased latency (Pearson's $r = -.50$, $p = .017$).

There were no statistically significant correlations between MMN amplitude or latency and any symptom factor nor were any significant differences seen in MMN parameters between diagnostic subgroups (all $p > .13$).

We expanded upon these results by investigating the spatial topography MMN amplitude and latency in HC and FEP (Fig. 2). In both HC and FEP, MMN amplitude was greatest in a broad region over the front of the scalp. Latency was longest in the occipital scalp in HC but this was not the case in FEP. We found a cluster of 63 channels over the front half of the scalp that had decreased MMN amplitude in FEP compared to HC (SnPM suprathreshold cluster test, $p = .039$). We also found a small cluster of 6 occipital electrodes that had longer latencies in HC compared to FEP (SnPM suprathreshold cluster test, $p = .047$).

3.2. Directional connectivity is decreased following deviant stimuli compared to standard stimuli and in FEP compared to HC

We selected four channels, located at opposite ends of the large amplitude cluster, to investigate long-range connectivity following stimulus presentation (Fig. 3a). We calculated the Granger causality between each pair of channels and the causal density across the network formed by the channels (Fig. 3b). For causal density, multivariate ANOVA showed effects of group ($F_{1,89} = 4.98$, $p = .028$) and stimulus type ($F_{1,89} = 30.68$, $p < .0001$) and no statistically significant interaction between group and stimulus ($F_{1,89} = 0.01$, $p = .92$). Post-hoc testing

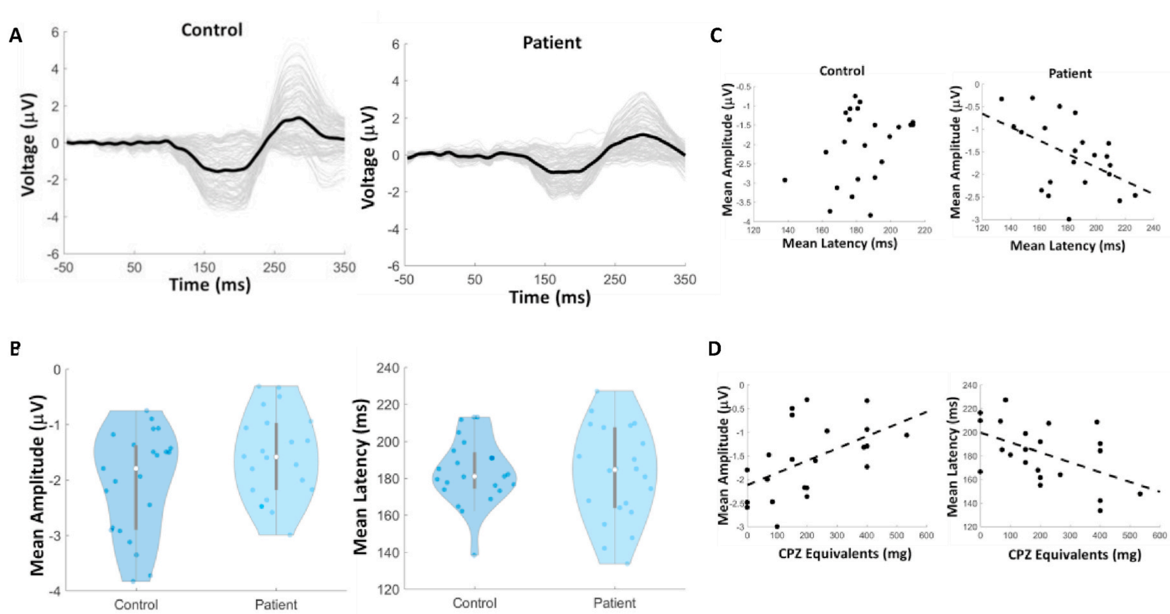

Fig. 1. Mismatch negativity amplitude and latency. A) Average difference waves for deviant – standard stimuli in all channels in controls and patients. The black traces are the average difference waves across all channels. B) Violin plots of the average MMN amplitude and latency across all channels in controls and patients. C) Scatterplots showing the relationship between mean amplitude and mean latency across all channels in controls and patients. Trendline indicates a statistically significant relationship in the patient group (Pearson's $r = -.49$, $p = .020$). D) Scatterplots showing the relationship between chlorpromazine equivalents and mean amplitude (Pearson's $r = .51$, $p = .015$) and latency (Pearson's $r = -.50$, $p = .017$) across all channels.

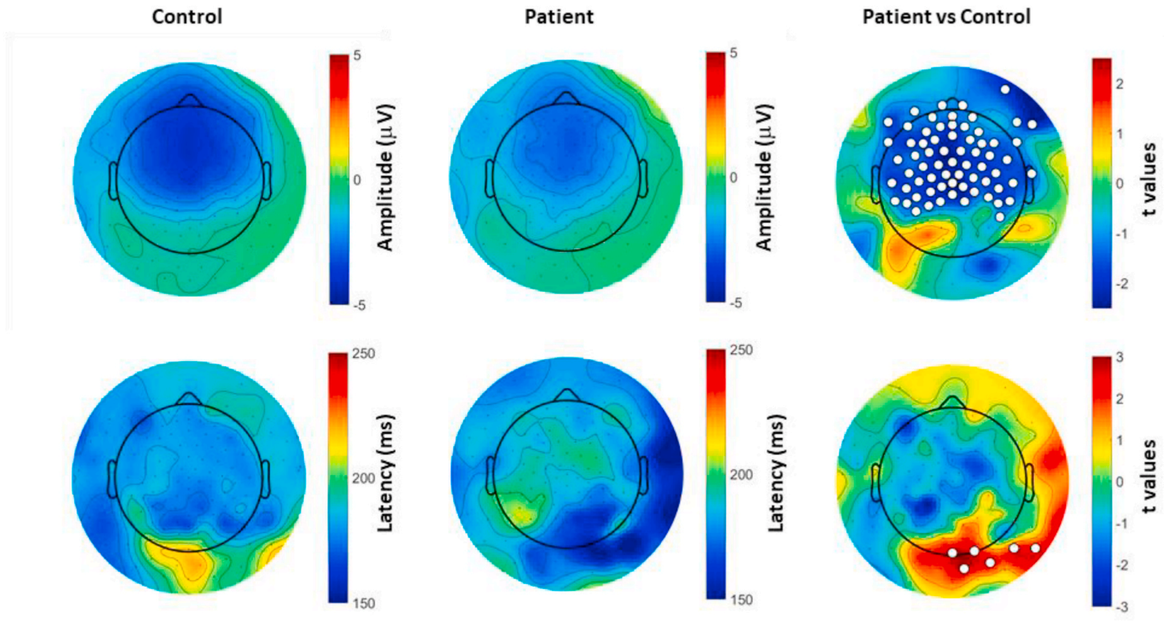

Fig. 2. Topography of aberrant mismatch negativity amplitude and latency in first-episode psychosis. The first column shows the topography of the mean MMN amplitude and latency in controls. The second column shows the topography of the mean MMN amplitude and latency in patients. The third column shows the comparison between patients and controls. The white dots indicate electrodes that statistically significantly different (SnPM suprathreshold cluster test, $p < .05$).

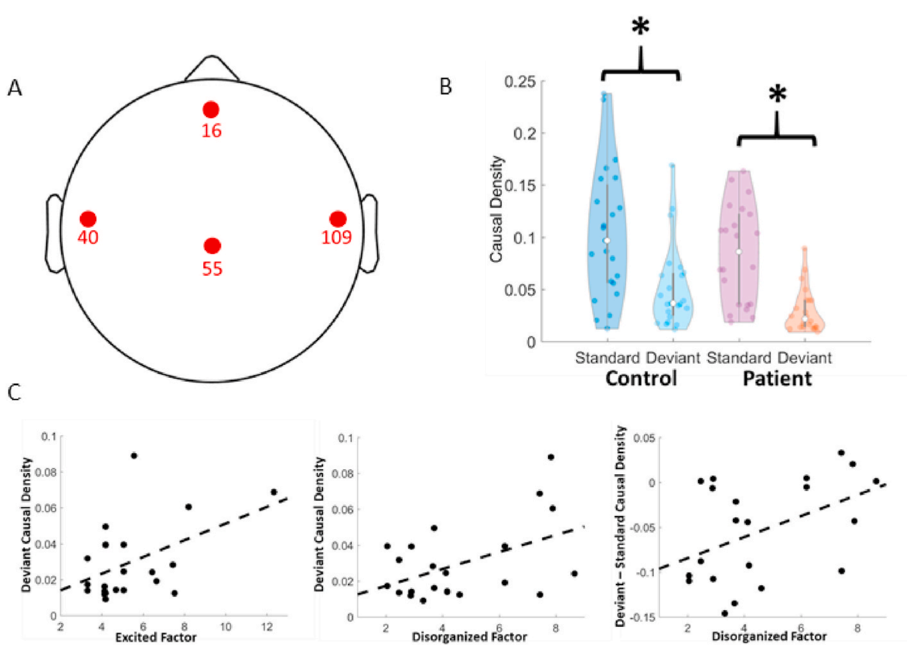

Fig. 3. Long-range causal density is higher during standard stimuli compared to deviant stimuli. A) Four spatially distinct electrodes from the cluster identified in Fig. 2 were chosen for the causal density analysis. B) Violin plots showing the mean causal density during standard and deviant stimuli in patients and controls. * $p < .000002$, Tukey's HSD test. C) Scatterplots showing the relationship between causal density during the deviant stimuli and Excited factor (Pearson's $r = .47$, $p = .028$) and Disorganized factor (Pearson's $r = 0.46$, $p = .030$ as well as between the difference between deviant and standard causal density and the Disorganized factor (Pearson's $r = 0.44$, $p = .038$).

(Tukey's HSD) showed that HC (0.08 SEM 0.007) had higher causal density than FEP (0.06 SEM 0.007, $p = .027$) and standard stimuli (0.09 SEM 0.007) had higher causal density than deviant stimuli (0.04 SEM 0.007, $p < .00001$). Furthermore, HC standard stimuli (0.10 SEM 0.01) had greater causal density than HC deviant stimuli (0.04 SEM 0.008, $p < .0001$) and FEP deviant stimuli (0.03 SEM 0.005, $p < .0001$). FEP standard stimuli (0.08 SEM 0.01) had greater causal density than FEP deviant stimuli ($p < .0001$). In an exploratory analysis, we tested whether there were associations between causal density and symptom scores (Fig. 3c). We found that causal density following deviant stimuli was associated with both the Disorganized (Pearson's $r = 0.47$, $p = .02$) and the Excited factor (Pearson's $r = 0.46$, $p = .03$). Patients who had greater Disorganization scores had smaller differences in causal density between deviant and standard stimuli (Pearson's $r = 0.44$, $p = .04$). Causal density was not associated with MMN amplitude or latency in either HC or FEP (all $p > .09$).

3.3. *Abnormal connectivity during deviant stimuli is associated with psychosis symptoms*

Next, we examined the topography of connectivity between the four electrodes (Fig. 4). Overall and consistent with the causal density findings, most connections had decreased magnitude in the deviant stimuli compared to standard stimuli. For each group and condition, we

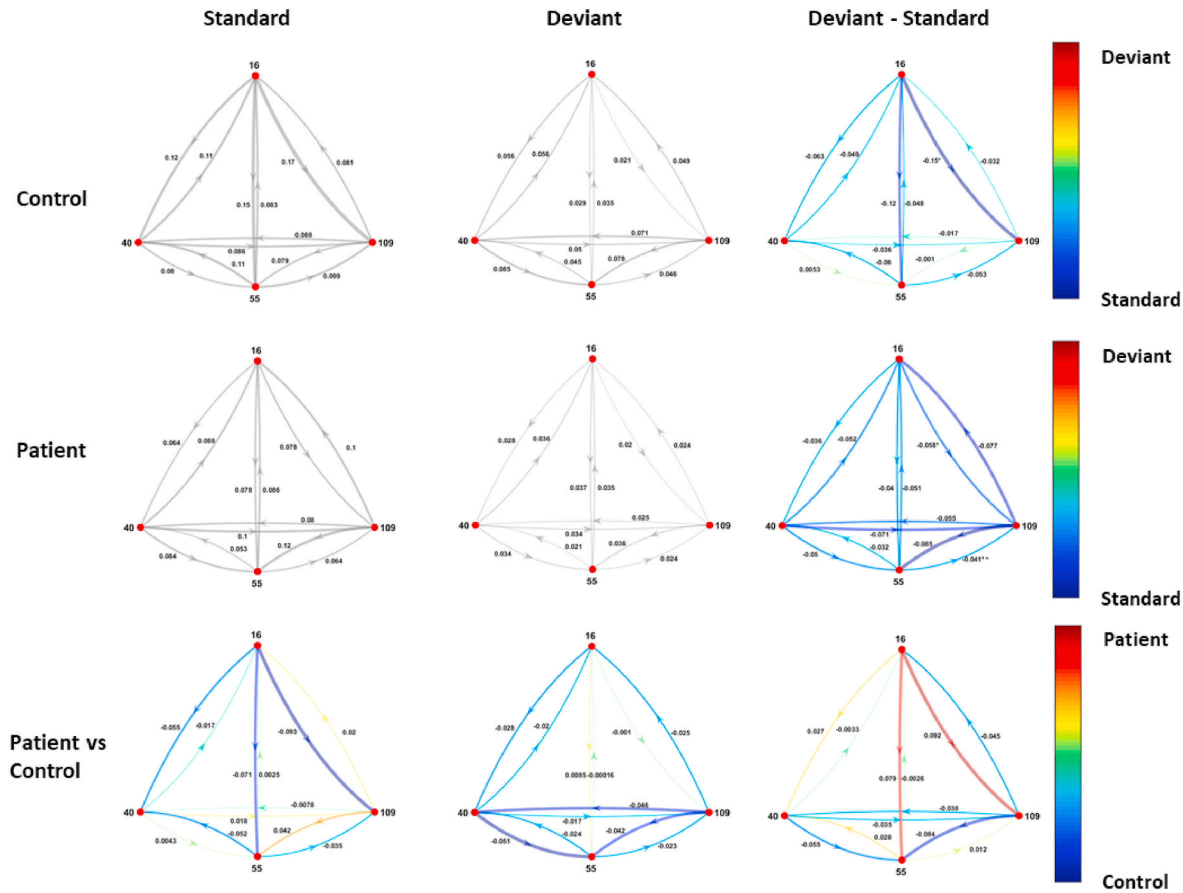

Fig. 4. Granger causality directed graphs. Top row shows the Granger causality (GC) values for the connections between the channels identified in Fig. 3a. The thickness of the line is proportional to the magnitude of the GC. The third column shows the difference between GC in the deviant vs the standard stimuli. Red connections are stronger in the deviant condition and blue connections are stronger in the standard condition. * p = .0002, SnPM test. Second row shows the GC values for the patient group. ** p = .01, SnPM test * p = .04, SnPM test. Third row shows the difference between patients and controls during standard stimuli, deviant stimuli, and the difference between deviant and standard stimuli. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

calculated an ANOVA to probe for topographic variation in connectivity strength. After correction for multiple comparisons, we found evidence of topographic variation in connection strength in the HC, deviant stimuli condition ($F_{11,242} = 3.6$, $p_{adjusted} = 0.0004$). Post-hoc testing showed that connectivity from channel 40 (left) to channel 55 (central) was greater than connectivity in both directions between channel 16 (frontal) and channel 55 (p = .0047,.022, respectively, Tukey's HSD) as well as greater than connectivity between frontal and channel 109 (right, p = .0004,Tukey's HSD). In the same condition, connectivity from right to central was greater than connectivity from frontal to central (p = .032) and from frontal to right (p = .004).

In order to assess patterns of directional connectivity following the different stimuli, we categorized each connection as front-to-back, back-to-front, left-to-right, and right-to-left. The front-to-back group consisted of the following connections: frontal-to-central, frontal-to-left, frontal-to-right, right-to-central, and left-to-central. The back-to-front group consisted of the opposite connections. The right-to-left group consisted of the following connections: right-to-frontal, right-to-left, right-to-central, central-to-left, and frontal-to-left. The left-to-right group consisted of the opposite connections. Some connections were assigned to more than one category. Bonferroni-corrected ANOVAs on group x direction showed evidence of a stimulus effect for all directions (all $p_{adjusted} < .0003$). In HC, we found that front-to-back connectivity was decreased following the deviant stimuli compared to the standard stimuli ($p_{adjusted} = .010$, paired *t*-test, Bonferroni corrected, Cohen's d = 0.95) as were back-to-front connectivity ($p_{adjusted} = .026$, paired *t*-test, Bonferroni corrected, Cohen's d = 0.68) and left-to-right connectivity ($p_{adjusted} = .011$, paired *t*-test, Bonferroni corrected, Cohen's d = 0.94). In contrast, in FEP, all directions of connectivity were decreased: front-to-back ($p_{adjusted} = .0005$, paired *t*-test, Bonferroni corrected, Cohen's d = 1.28), back-to-front ($p_{adjusted} = .002$, paired *t*-test, Bonferroni corrected, Cohen's d = 1.14), right-to-left ($p_{adjusted} = .0034$, paired *t*-test, Bonferroni corrected, Cohen's d = 0.95), and left-to-right ($p_{adjusted} = .009$, paired *t*-test, Bonferroni corrected, Cohen's d = 1.10). Comparing HC to FEP, we found that during deviant stimuli right-to-left connectivity was higher in HC compared to FEP ($p_{adjusted} = .04$, unpaired *t*-test, Bonferroni corrected, Cohen's d = 0.65). There were no other statistically different connectivities between HC and FEP. Right-to-left connectivity was not correlated with CPZ equivalents (Pearson's r = -.08, p = .72) but was associated with Excited (Pearson's r = 0.44, p = .04) and Disorganized (Pearson's r = 0.48, p = .023).

We then compared how connectivity between pairs of channels differed between the standard and deviant stimuli. We found that in both HC and FEP, connectivity from frontal to right was decreased following deviant stimuli compared to standard stimuli (SnPM, p = .0002 and p = .01, respectively). In FEP, connectivity from central to right was also statistically significantly decreased following deviant stimuli compared to standard stimuli (SnPM, p = .04). There were no statistically significant differences between HC and FEP. There were no statistically significant associations between frontal to right and center to right connectivity and symptom scores or CPZ.

For each channel, we calculated net information flow for each channel by subtracting the inward Granger causality from the outward Granger causality (Fig. 5). There were no differences in flow in any channel between HC and FEP. We then calculated whether there were differences in flow between stimuli type. In HC, we found a statistically significant increase in flow in the right channel in deviant stimuli compared to standard stimuli ($p_{adjusted}$ = .016, paired *t*-test, Bonferroni adjusted).

4. Discussion

We used auditory duration mismatch negativity to probe long-range, directional cortical connectivity in HC and FEP using high-density EEG. Consistent with past work, we found that MMN amplitudes were largest over the fronto-central region of the scalp in both HC and FEP and that MMN amplitude was broadly decreased over this same region in FEP compared to HC. We also found that MMN latency was decreased in the occipital scalp. Within a four-channel network consisting of frontal, central, left, and right channels we found increased causal density in HC compared to FEP and increased causal density following standard stimuli compared to deviant stimuli. Relatively high causal density following deviant stimuli was associated with higher symptom scores for disorganization and excitement. Connectivity involving the right side of the scalp was modulated by the stimulus type. Aberrant modulation of connectivity to and from the right scalp in response to stimulus type was noted in FEP and was associated with greater symptom burden. In HC, the right side of the scalp becomes a relative information source following deviant stimuli compared to standard stimuli, however this effect was not observed in FEP. Taken together, our results suggest that bottom-up processing abnormalities in long-distance cortical circuits contribute to the well-known MMN abnormalities observed in psychotic disorders.

5. Mismatch negativity and psychosis

Decreased amplitude of duration MMN has been repeatedly reported in patients with schizophrenia. However, some evidence suggests that the magnitude of the effect is larger in chronic populations compared to first-episode or ultra-high risk populations (Atkinson et al., 2012; Haigh et al., 2017; Salisbury et al., 2002). Furthermore, patients with bipolar disorder have diminished MMN compared to healthy controls, but not as diminished as patients with schizophrenia (Jahshan et al., 2012; Umbricht et al., 2003). Our transdiagnostic sample of patients with first-episode psychosis compliments this literature and provides further evidence for attenuated MMN in this population. The relationship between medication and MMN is not entirely clear. Small, longitudinal studies of risperidone and olanzapine found no effect of these medications on MMN (Iwanami et al., 2001; Korostenskaja et al., 2005). Another study showed that clozapine dose was negatively correlated with MMN amplitude (Horton et al., 2011). We found that MMN amplitude was negatively correlated with antipsychotic medication dose, suggesting either a direct medication effect on MMN or, more likely, confounding by severity where more severely affected patients have greater MMN abnormalities and are on higher doses of medication. When interpreting out results in the context of this literature, note that we measured average MMN amplitude across the entire scalp, rather than restricting our analysis to a single channel or small set of channels. We chose this analysis because our results suggest that mismatch negativity deficits can be detected broadly over the scalp rather than only in a small subset of channels. While decreased MMN latency has been previously reported in schizophrenia, previous studies have not shown consistent effects of antipsychotic medication on MMN latency (Kärgel et al., 2014; Korostenskaja et al., 2005; Umbricht et al, 1998, 1999). The spatial considerations mentioned above and the fact that our sample was transdiagnostic and early course may explain the difference between our results and this literature.

Our directed connectivity results are partially consistent with a recent report that patients with chronic schizophrenia have decreased inter-hemispheric effective connectivity during MMN, however, we found increased connectivity from front to back in patients while the chronic population had increased connectivity from back to front (Koshiyama et al., 2020). This suggests that different patterns of dysconnectivity may underlie MMN deficits in early vs chronic schizophrenia perhaps reflecting the progression of disease.

Predictive coding, has been hypothesized to contribute to MMN generation (Michie et al., 2016). In predictive coding models, the brain uses information from previous sensory inputs to generate models to predict future sensory inputs. Prediction errors occur when there is a discrepancy between the predicted and actual input. These errors are used to update and refine the predictive model while also allocating neural resources to the unexpected stimuli. In the predictive coding framework, bottom-up inputs from sensory cortices are inhibited by top-down inputs from frontal cortex (Kenemans and Kähkönen, 2011). Dynamic causal modelling studies of MMN indicate that such hierarchical models are best able to recapitulate the MMN response (Garrido et al., 2009; Kirihara et al., 2020). Furthermore, NDMA receptors, which are known to modulate MMN, provide a biologically plausible mechanism for generation and maintenance of predictive coding models (Michie et al., 2016). In addition, NMDA receptor antagonists decrease the MMN response (Heekeren et al., 2008) with smaller MMN amplitudes being associated with schizophrenia-like negative symptoms (Thiebes et al., 2017). Our findings are consistent with disrupted NMDA receptor function in our transdiagnostic sample.

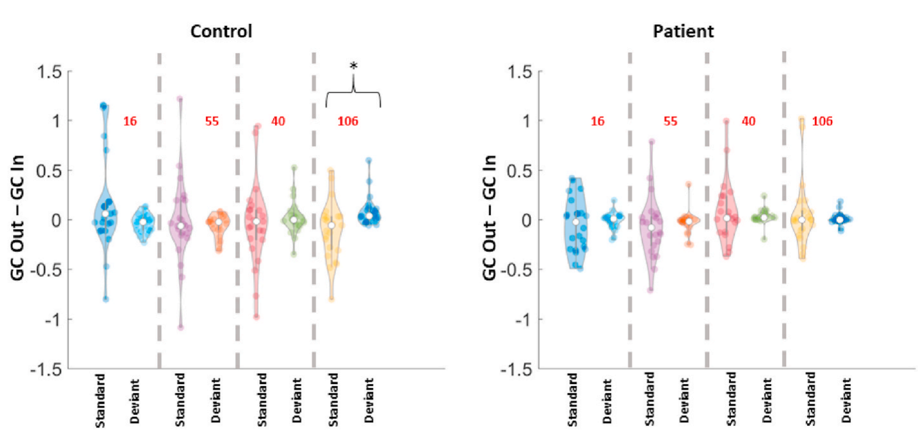

Fig. 5. Flow (Granger causality out – Granger causality in). Violin plots showing the difference between GC out and GC in in each of the four channels during deviant and standard stimuli in the control and patient groups. * $p_{adjusted}$ = .032, paired *t*-test, Bonferroni correction.

6. Dysconnectivity and top-down vs bottom-up processing

Auditory MMN is largely driven by neural activity in the auditory cortex but is modulated by the frontal cortex and the thalamus (Lakatos et al., 2020). This suggests that MMN relies on bottom-up processing but can be impacted by top-down inputs. Given this reliance on both local cortical microcircuits and longer-range cortico-thalamo-cortical networks, it is unsurprising that psychotic disorders, which are characterized by myriad forms of cortical dysconnectivity, are accompanied by decreased MMN. Dysconnectivity in schizophrenia is complex with research findings depending upon brain regions, behavioural state, illness duration, medication status, and connectivity measures (Anticevic et al, 2014a, 2014b; Di Lorenzo et al., 2015). Causal density analysis provides a way to quantify dysconnectivity in the brain. Our causal density results suggest that abnormally complex patterns of connectivity during deviant stimuli and failure to attenuate the complexity of the pattern of connectivity, were associated with psychotic symptoms. Furthermore, we report a general decrease in connectivity within the front half of the head in FEP compared to HC. This is consistent with tractography data showing broadly decreased anisotropy in schizophrenia (Kelly et al., 2018). We found that, following deviant stimuli, long-range Granger causality across the scalp is decreased compared to following standard stimuli. Therefore, in the deviant condition, EEG signals are more driven by local activity than by long range inputs. The relative freeing of local micro-circuits from global modulation is consistent with a greater role of bottom-up vs top-down processing following the deviant stimuli in both HC and FEP. Connectivity differences between HC and FEP were driven by smaller front-to-midline connectivity following standard stimuli and smaller horizontal connectivity following deviant stimuli. The aberrant persistent connectivity from frontal regions to midline and right temporal regions in FEP compared to HC is consistent with a model in which MMN deficits in FEP are the result of failure of inappropriately preserved top-down processing and/or diminished bottom-up processing following a novel stimulus in patients. Increased top-down processing in schizophrenia has been widely reported and may be related to hallucinations (Aleman et al., 2003). Impaired bottom-up processing has also been reported in some auditory paradigms as well (Adcock et al., 2009). Our work builds on these reports by providing evidence for specific failures to modulate the balance between these processes in FEP and relates this failure to thought disorder. Future work should incorporate visual paradigms to test whether these processing abnormalities exist in other sensory modalities and other regions of the brain such as the occipital cortex. Future work should also identify whether there is evidence for dysconnectivity, and therefore top-down vs bottom-up imbalance, in pitch variant MMN paradigms where amplitude deficits are not present in first episode psychosis but are present in patients with chronic psychosis. This would provide clarity as to whether inappropriate modulation of processing is non-specific global phenomenon or whether there are varied, more subtle, deficits in modulation which rely on distinct neural circuits and can be interrogated by specific stimulus parameters.

7. Limitations

Our study has several limitations that should be considered when interpreting our results. The cross-sectional design of this study limits our ability to investigate the relationship between abnormal MMN and disease progression or medication exposure. Further work could investigate whether the administration of antipsychotic medications to healthy controls produces reliable changes in MMN. In addition, our transdiagnostic sample makes direct comparison to previous studies restricted to schizophrenia non-trivial. Inferences from EEG recordings to cortical activity are limited by the complicated relationship between neural activity and detectable scalp voltages. Nonetheless, the data presented in this study are broadly consistent with past work showing diminished MMN in individuals with psychotic disorders. We expand on this past work by identifying specific patterns of dysconnectivity during the MMN paradigm in patients with early course psychosis and linking this dysconnectivity to symptom burden. We note EEG data is easy to collect compared to other imaging methods and that connectivity measures described in this study are relatively straightforward to calculate suggesting that GC may be a widely accessible clinical measure.

Author statement

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